

50 Power BI Interview Questions with Answers

These questions cover the most commonly asked topics in Power BI interviews, from beginner to advanced.

Power BI Basics

1. What is Power BI?

Power BI is a business intelligence and data visualization tool developed by Microsoft. It helps organizations analyze data, create interactive reports, and share insights.

2. What are the main components of Power BI?

- Power BI Desktop
 - Power BI Service
 - Power BI Mobile
 - Power BI Gateway
 - Power BI Report Server
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3. What is the difference between Power BI Desktop and Power BI Service?

Power BI Desktop is used for creating reports, while Power BI Service is used for publishing, sharing, and collaborating online.

4. What are the building blocks of Power BI?

- Visualizations
 - Datasets
 - Reports
 - Dashboards
 - Tiles
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5. What is a dashboard?

A dashboard is a single-page canvas that displays key metrics and visuals from one or more reports.

Data Import and Connectivity

6. What data sources can Power BI connect to?

Power BI supports Excel, SQL Server, Oracle, MySQL, SharePoint, Azure, Google Analytics, and many more.

7. What is Import Mode?

Data is imported into Power BI's memory, providing faster performance.

8. What is DirectQuery?

DirectQuery connects directly to the data source without importing data.

9. Difference between Import and DirectQuery?

Import is faster but consumes memory. DirectQuery provides real-time data but may be slower.

10. What is Live Connection?

It connects directly to Analysis Services or an existing Power BI dataset.

Power Query

11. What is Power Query?

Power Query is the ETL tool used for data extraction, transformation, and loading.

12. What language does Power Query use?

Power Query uses M Language.

13. What is Query Folding?

Query Folding pushes transformations back to the source system for better performance.

14. Difference between Merge and Append?

Merge combines tables horizontally using keys. Append stacks tables vertically.

15. What are Applied Steps?

They are the sequence of transformations performed in Power Query.

Data Modeling

16. What is a data model?

A data model defines relationships between multiple tables.

17. What is Star Schema?

A central fact table connected to multiple dimension tables.

18. Difference between Star Schema and Snowflake Schema?

Star Schema is denormalized; Snowflake Schema is normalized.

19. What is Cardinality?

It defines the relationship type between tables.

20. Types of Relationships in Power BI?

- One-to-One
 - One-to-Many
 - Many-to-One
 - Many-to-Many
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DAX

21. What is DAX?

DAX (Data Analysis Expressions) is the formula language used in Power BI.

22. Difference between Measure and Calculated Column?

Measure is calculated dynamically; Calculated Column is stored in the model.

23. What is CALCULATE()?

CALCULATE modifies the filter context of a calculation.

Total Sales = CALCULATE(SUM(Sales[Amount]))

24. What is FILTER()?

FILTER returns a subset of a table based on conditions.

25. What is ALL()?

ALL removes filters from a table or column.

26. What is RELATED()?

RELATED fetches data from a related table.

27. What is SUMX()?

SUMX iterates through each row and performs calculations.

28. Difference between SUM and SUMX?

SUM adds a single column; SUMX evaluates an expression row by row.

29. What is Context in DAX?

- Row Context

- Filter Context
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30. What is Time Intelligence?

Functions like TOTALYTD, SAMEPERIODLASTYEAR, and DATEADD.

Visualizations

31. What is a Slicer?

A slicer is a visual filter for reports.

32. Difference between Filter and Slicer?

Filters are pane-based; slicers are report visuals.

33. What is Drill Down?

It allows navigation from summary to detailed data.

34. What is Drill Through?

It navigates to another report page with contextual filtering.

35. What are Bookmarks?

Bookmarks save report states for storytelling and navigation.

Security

36. What is Row-Level Security (RLS)?

RLS restricts data access for different users.

37. Types of RLS?

- Static RLS
- Dynamic RLS

Power BI Service

38. How do you publish a report?

Use the Publish button in Power BI Desktop.

39. What is a Workspace?

A collaborative environment for reports and dashboards.

40. What is a Gateway?

A gateway securely connects on-premises data sources to Power BI Service.

41. What is Scheduled Refresh?

Automatic dataset refresh at configured intervals.

42. What is a Dataflow?

A reusable ETL process in Power BI Service.

Performance Optimization

43. How do you improve Power BI performance?

- Use Star Schema
 - Remove unnecessary columns
 - Optimize DAX
 - Enable query folding
 - Use aggregations
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44. What is Incremental Refresh?

Only newly added or changed data is refreshed.

Advanced Concepts

45. What is Composite Model?

A model combining Import and DirectQuery.

46. What is Bidirectional Filtering?

Filters propagate in both directions between related tables.

47. What are Aggregations?

Pre-summarized tables for faster querying.

48. What is Power BI Embedded?

It allows embedding Power BI reports into custom applications.

Scenario-Based Questions

49. How would you optimize a slow report?

- Reduce visuals
 - Optimize DAX
 - Use Import mode
 - Remove unused columns
 - Create summary tables
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50. How do you lock a visual in Power BI?

- Go to the **View** tab.
- Enable **Lock Objects**.
- This prevents accidental movement or resizing